

# Dipole-engineered co-adsorbed self-assembled monolayers boost Cs-based inorganic Pb-Sn perovskite tandem efficiency beyond 20%

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Self-assembled monolayers (SAMs) have emerged as pivotal components for achieving high-performance perovskite solar cells (PSCs). However, most high-efficiency PSCs employing SAMs utilize Pb-based wide-bandgap (WBG) perovskites, whereas their application in tandem-compatible Pb-Sn mixed narrow-bandgap (NBG) perovskites remains scarcely documented. Herein, high-performance PSCs with Cs-based inorganic Pb-Sn absorbers are fabricated through dipole engineering via co-assembling 4-[(4-aminophenyl)diazetyl] benzenesulfonic acid (4-ASA) with Me-4PACz on  $\text{NiO}_x$ . We demonstrated that 4-ASA induces intermolecular interactions with Me-4-PACz, inhibiting SAM aggregation and micelle formation. This enables the formation of co-SAMs with oriented dipole moments directed from  $\text{NiO}_x$  to the perovskite, thereby effectively optimizing the energy-level alignment at the buried interface. Moreover, the co-SAMs interlayer exhibits a superior hydrophilic surface, serving as an ideal nucleation template for perovskite deposition, contributing to perovskite films with increased grain boundary groove (GBG) angles, suppressed interfacial  $\text{Sn}^{2+}$  oxidation, and mitigated residual tensile stress. As a result, the NBG Pb-Sn-based inorganic PSCs achieve a scanned power conversion efficiency (PCE) of 17.82% with outstanding ultraviolet irradiation and operational stability. Furthermore, by integrating NBG (1.36 eV,  $\text{CsPb}_{0.6}\text{Sn}_{0.4}\text{I}_3$ ) and WBG (1.74 eV,  $\text{CsPb}_{0.7}\text{Sn}_{0.3}\text{IBr}_2$ ) subcells, we demonstrate the first four-terminal (4 T) perovskite tandem device in which both subcells are based on Cs-based Pb-Sn mixed inorganic absorbers, achieving a total scanned PCE of 20.36%.

In recent years, self-assembled monolayers (SAMs) have emerged as versatile and effective hole transport layers (HTLs) in inverted perovskite solar cells (PSCs) and perovskite-based tandem solar cells (TSCs), owing to their solution processability, low parasitic absorption, precise energy level modulation, and ability to passivate defects in both the overlying perovskite films and the underlying oxide substrates<sup>1–4</sup>.

Despite their potential, achieving densely packed SAMs on indium tin oxide (ITO) substrates remains challenging, resulting in poorly controlled interfacial properties. This limitation stems primarily from two factors. First, the inherently weak adsorption affinity between SAMs and bare ITO surfaces drives desorption and molecular rearrangement during subsequent perovskite deposition<sup>5,6</sup>. Second, the

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amphiphilic nature of SAMs leads to the formation of micellar aggregates in solution<sup>7,8</sup>. These effects promote the formation of clusters on the ITO substrate, leading to incomplete surface coverage<sup>9</sup>. A highly effective strategy to mitigate this issue is to construct bilayer HTLs by depositing SAMs onto NiO<sub>x</sub> substrates<sup>10</sup>. For instance, Lv et al. pre-adsorbed arginine (Arg) onto NiO<sub>x</sub> substrates to suppress SAMs self-aggregation through ionic interactions and steric hindrance, enabling PSCs with efficiency of 26.67% (0.049 cm<sup>2</sup>) and minimodule efficiency of 21.05% (64.68 cm<sup>2</sup>)<sup>11</sup>. However, most high-efficiency PSCs employing SAMs-modified NiO<sub>x</sub> HTLs utilize Pb-based organic-inorganic hybrid perovskites, whereas their application in Cs-based inorganic Pb-Sn PSCs has been rarely documented. This is particularly notable given that Pb-Sn-based inorganic perovskites have recently emerged as a highly promising material class for tandem photovoltaic applications. Relative to their pure Pb-based organic-inorganic hybrid counterparts, Pb-Sn mixed inorganic perovskites exhibit multiple compelling advantages, including lower fabrication costs, higher intrinsic thermal stability due to the absence of volatile organic cations, reduced Pb content, and readily tunable band gaps across favorable energy ranges<sup>12–14</sup>.

Such a research gap primarily originates from the energy level misalignment between NiO<sub>x</sub> and Pb-Sn perovskites. Specifically, NiO<sub>x</sub> typically exhibits a valence band maximum (VBM) of approximately –5.33 eV relative to vacuum<sup>15</sup>. Pb-based organic-inorganic hybrid perovskites (FAPbI<sub>3</sub>) demonstrate a deeper VBM of –5.45 eV, which gives rise to a gentle downward energy offset of 0.12 eV at the FAPbI<sub>3</sub>/NiO<sub>x</sub> interface and thus facilitates efficient hole extraction<sup>16</sup>. In contrast, narrow-bandgap (NBG) Pb-Sn-based inorganic perovskites such as CsPb<sub>0.6</sub>Sn<sub>0.4</sub>I<sub>3</sub> generally possess a shallower VBM of about –5.0 eV owing to Sn-s/I-p orbital hybridization, which creates an upward energy barrier of 0.33 eV at the CsPb<sub>0.6</sub>Sn<sub>0.4</sub>I<sub>3</sub>/NiO<sub>x</sub><sup>17,18</sup>. This pronounced interfacial mismatch severely hinders hole extraction, enhances charge recombination, and ultimately causes significant open-circuit voltage (V<sub>OC</sub>) losses and deteriorates overall device performance. Although SAMs deposited on NiO<sub>x</sub> can induce dipole moments oriented from NiO<sub>x</sub> toward the perovskite layer, an effect that potentially elevates the VBM of NiO<sub>x</sub>, the commonly used SAMs molecules such as Me-2PACz and Me-4PACz exhibit relatively small dipole moments of ~2 Debye<sup>19,20</sup>. This limited dipole strength is insufficient to compensate for the substantial energy level offset at NiO<sub>x</sub>/Pb-Sn perovskite interfaces. Furthermore, Me-2PACz and Me-4PACz molecules exhibit severe aggregation during deposition, resulting in disordered dipole orientations and incomplete surface coverage. Such incomplete coverage leaves exposed NiO<sub>x</sub> regions directly in contact with the perovskite layer, which triggers unfavorable Sn<sup>2+</sup> to Sn<sup>4+</sup> redox reactions and generates abundant non-radiative recombination centers at the NiO<sub>x</sub>/Pb-Sn perovskite interface<sup>21</sup>.

Intriguingly, in our previous work, we pioneered 4-aminobenzenesulfonic acid (4-ABSA) as a multifunctional dipole-engineered interlayer to mitigate the energy level misalignment at the NiO<sub>x</sub>/Pb-Sn perovskite interface. This molecular design generates a strong interfacial dipole moment (7.05 D) directed from NiO<sub>x</sub> to perovskite, upshifting the VBM of NiO<sub>x</sub> by 0.29 eV. Consequently, the energy offset at the NiO<sub>x</sub>/Pb-Sn-based inorganic perovskite was reduced from 0.30 eV to 0.01 eV, contributing to a significant enhancement in charge extraction efficiency<sup>22</sup>. However, 4-ABSA contains limited charge-transport moieties, such as extended  $\pi$ -systems, resulting in moderate hole mobility performance. Additionally, its zwitterionic structure induces notable intramolecular charge confinement, partially interrupting  $\pi$ -conjugation pathways and reducing the strength of intermolecular  $\pi$ - $\pi$  stacking interactions. Moreover, the relatively large molecular dipole moment causes electrostatic interactions to prevail over  $\pi$ - $\pi$  stacking between molecules. As a result, the growth of closely packed, continuous surface films becomes somewhat constrained. Therefore, it is imperative to design and

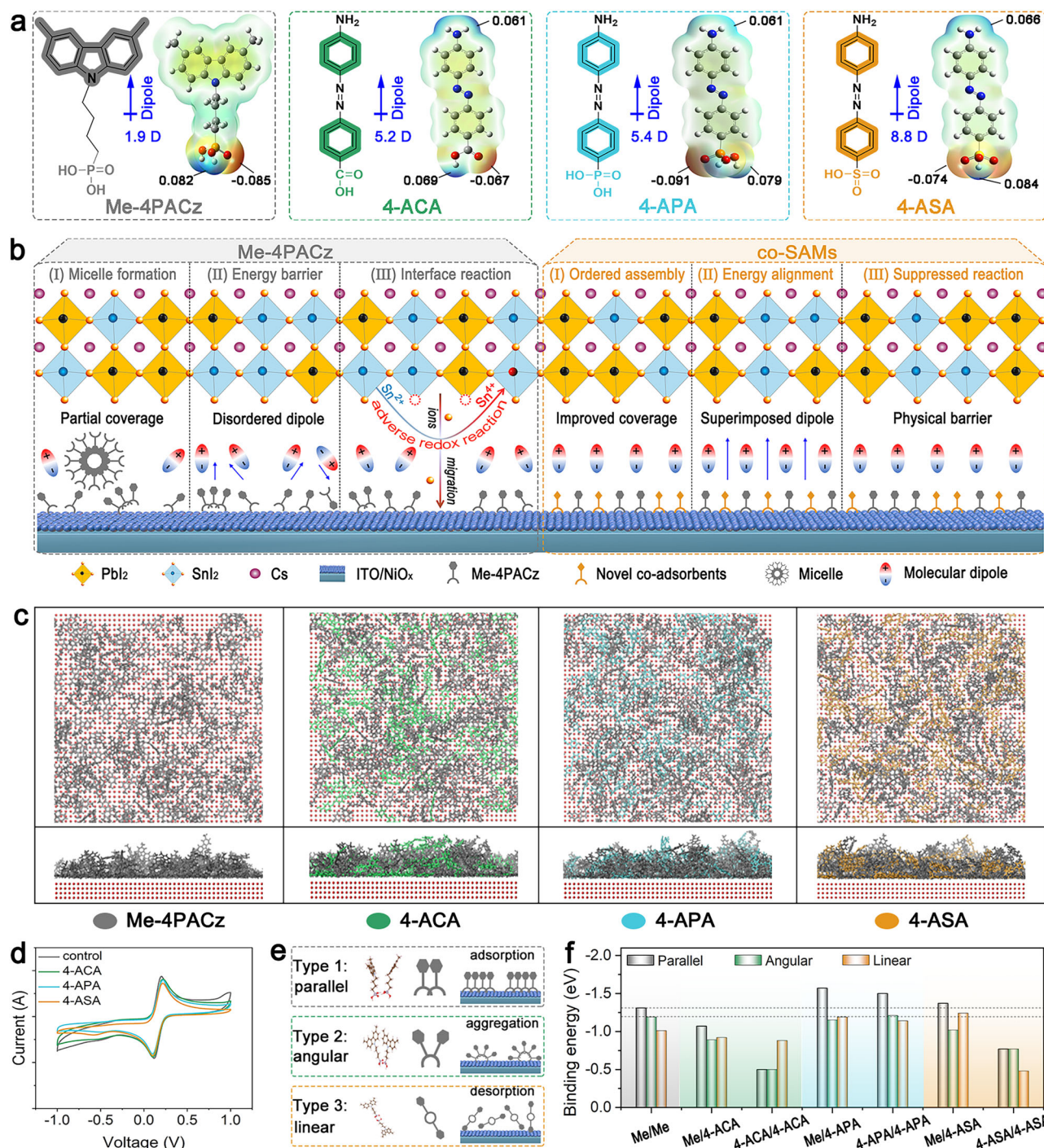
synthesize novel SAMs featuring superior hole mobility and high dipole moment. By exploiting the additive dipole superimposition effect<sup>23</sup>, co-assembling these novel SAMs with conventional counterparts (e.g., Me-4PACz) to construct synergistic mixed SAMs on NiO<sub>x</sub> is expected to promote the development of high-performance Pb-Sn-based inorganic PSCs.

Herein, we investigated dipole engineering through co-assembling azobenzene derivatives, specifically 4-[(4-aminophenyl) diazenyl]phenyl analogs bearing carboxylic acid (4-ACA), phosphonic acid (4-APA), and sulfonic acid (4-ASA) anchoring groups, with Me-4PACz, to construct NiO<sub>x</sub>/co-SAMs bilayer HTLs for high-performance Pb-Sn-based inorganic PSCs. Molecular dynamics (MD) simulations and experimental results revealed kinetically controlled competitive adsorption between these molecules and Me-4PACz. Optimal co-adsorption was achieved by employing high-dipole (8.8 D) 4-ASA as a co-adsorbent with Me-4PACz. The resulting co-SAM layers exhibited improved surface coverage and a superimposed dipole moment directed from NiO<sub>x</sub> to perovskite, which effectively elevates the VBM of NiO<sub>x</sub> and reduces the interfacial energy offset to 0.02 eV. For devices, 4-ASA-based co-SAMs-modified substrate increases perovskite grain boundary groove (GBG) angles, suppresses interfacial Sn<sup>2+</sup> oxidation, and mitigates residual tensile stress, contributing to a flattened perovskite buried interface and improved crystallinity. These synergistic effects yield a remarkable scanned PCE of 17.82% for single-junction NBG Pb-Sn-based inorganic PSCs. Furthermore, by integrating NBG (1.36 eV, CsPb<sub>0.6</sub>Sn<sub>0.4</sub>I<sub>3</sub>) and WBG (1.74 eV, CsPb<sub>0.7</sub>Sn<sub>0.3</sub>Br<sub>2</sub>) subcells, we demonstrate the first four-terminal (4 T) perovskite tandem device in which both the top and bottom subcells utilize Cs-based Pb-Sn mixed inorganic perovskite absorbers, yielding a total scanned PCE of 20.36%.

## Results and Discussion

### Competitive adsorption of co-SAMs

To realize dipole-engineered co-SAMs, a series of 4-[(4-aminophenyl) diazenyl]phenyl derivatives featuring distinct anchoring groups (carboxylic acid (–COOH), phosphonic acid (–PO<sub>3</sub>H<sub>2</sub>), and sulfonic acid (–SO<sub>3</sub>H)) were rationally designed. These three anchoring groups differ remarkably in acidity, charge delocalization, and coordination geometry, which govern NiO<sub>x</sub> substrate adsorption and intermolecular self-aggregation. Details of molecular synthesis and characterization are provided in the supplementary materials (Supplementary Fig. 1–3). Figure 1a presents the chemical structures and electrostatic potential (ESP) profiles of Me-4PACz and tailored molecules (4-ACA, 4-APA, and 4-ASA), where electron-withdrawing regions (negative red areas) correspond to the anchoring groups. According to density functional theory (DFT) calculations (Supplementary Fig. 4), 4-ACA, 4-APA, and 4-ASA exhibit molecular dipoles of 5.2 D, 5.4 D, and 8.8 D, respectively, significantly larger than that of Me-4PACz (1.9 D). Figure 1b schematically illustrates the interfacial modulation mechanisms of Me-4PACz and co-SAMs at the NiO<sub>x</sub>/Pb-Sn perovskite interface. Pure Me-4PACz tends to form micelles and aggregates, yielding partial SAMs coverage of the substrates. Additionally, the weak and disordered dipoles derived from Me-4PACz are insufficient to compensate for the substantial energy level offset at the NiO<sub>x</sub>/Pb-Sn perovskite interface, thereby creating an energy barrier to charge-carrier extraction. Moreover, the exposed NiO<sub>x</sub> triggers unfavorable Sn<sup>2+</sup> to Sn<sup>4+</sup> redox reactions and generates abundant non-radiative recombination centers, thus deteriorating device performance. In contrast, co-adsorbed tailored molecules are expected to form co-SAMs with improved surface coverage via competitive adsorption. Their strong superimposed molecular dipoles enable favorable energy-level alignment for charge-carrier extraction<sup>24,25</sup>. More importantly, the enhanced coverage of the co-SAMs serves as a physical barrier to reduce direct contact sites between NiO<sub>x</sub> and the Pb-Sn perovskite, thus suppressing the notorious interface redox reaction.



**Fig. 1 | Molecular interaction of co-SAMs. a** Molecular structures and electrostatic potential profiles of Me-4PACz, 4-ACA, 4-APA, and 4-ASA. **b** Schematic illustration of the functions of Me-4PACz and co-SAMs at the NiO<sub>x</sub>/perovskite interface. **c** Top and side views of the adsorption equilibrium of different co-SAMs on ITO/NiO<sub>x</sub>.

surface simulated by molecular dynamics. **d** CV curves of different samples. **e** DFT optimized dimer structures of acid-acid interactions between Me-4PACz and tailored molecules. **f** Binding energies of parallel, angular, and linear configurations of acid-acid interactions.

To verify the proposed mechanisms, the binding modes of the tailored molecules with NiO<sub>x</sub> were firstly investigated. The results indicate that these molecules preferentially adsorbed on the NiO<sub>x</sub> surface by anchoring through their acidic groups (Supplementary Fig. 5), whereas amino groups (–NH<sub>2</sub>) likely interact with the perovskite. To investigate the interactions between the tailored molecules and Me-4PACz during the co-adsorption process, molecular dynamics (MD) simulations were performed at 300 K using mixtures of Me-4PACz with 4-ACA, 4-APA, or 4-ASA at an experimentally relevant

molar ratio of 3:2. The simulations were conducted on a 10 nm × 10 nm NiO<sub>x</sub> surface for 1 ns, and the final MD snapshots are shown in Fig. 1c. The initial surface coverage of the constructed models ranged from 22.8% to 28.8%. Upon completion of the 1 ns simulation, Me-4PACz exhibited significant aggregation, yielding a surface coverage of 0.64. In contrast, the introduction of co-adsorbents increased substrate coverage to 0.73, 0.76, and 0.78 for Me-4PACz/4-ACA, Me-4PACz/4-APA, and Me-4PACz/4-ASA, respectively, indicating that these molecules effectively dispersed and adsorbed onto the NiO<sub>x</sub> surface during

the co-adsorption process. To further study the surface coverage of the different systems, cyclic voltammetry (CV) measurements were performed.  $\text{SnI}_2$  (20 mM) was added to a 0.5 M KCl electrolyte at pH = 5.5 to serve as a probe for the degree of electrical contact between the sample surface and the electrolyte. For clarity, the Me-4PACz-only system is designated as the control, while the mixed systems are denoted as 4-ACA, 4-APA, and 4-ASA, respectively. As shown in Fig. 1d, the CV curves of 4-ACA, 4-APA, and 4-ASA are overall narrower than that of the control. The features observed in the CV response of  $\text{NiO}_x$  indicate that its surface is electrochemically active and that molecular attachment to these surface sites can substantially suppress its electrochemical activity<sup>6</sup>. In this case, the reduced current response of the co-SAMs-modified systems is mainly attributed to the passivation or blocking of electrochemically accessible  $\text{NiO}_x$  sites, suggesting improved surface coverage. Among all the samples, the 4-ASA co-SAMs system exhibits the lowest current response, suggesting the lowest density of exposed electrochemically active sites and agreeing well with the MD simulations.

The enhanced surface coverage stems primarily from competitive adsorption between Me-4PACz and tailored molecules during co-adsorption. Binding energies of intermolecular interactions, including amino-amino, amino-acid, and acid-acid interactions, were systematically investigated (Supplementary Fig. 6–8), finding that acid-acid interactions are the strongest. To further decipher the competitive adsorption mechanism, parallel, angular, and linear configurations of acid-acid interactions were analyzed (Fig. 1e). Specifically, parallel configurations facilitate the formation of ordered SAMs through homogeneous acid anchoring. Angular configurations promote cluster formation, potentially leading to SAMs aggregations. While, linear configurations hinder stable substrate binding because they lack exposed acid anchoring groups. As shown in Fig. 1f, in the Me-4PACz-only system, the parallel Me/Me dimer (Me denotes Me-4PACz) exhibits the lowest binding energy (−1.31 eV), followed by angular (−1.19 eV) and linear (−1.01 eV) configurations. This indicates that despite Me-4PACz spontaneously forming micelles in polar alcohols<sup>12</sup>, most molecules adsorb on the substrate, while the remainder mainly aggregate through angular interactions. In the Me/4-ACA system, the localized negative charge of the −COOH groups generates strong dipole-dipole attraction, making Me/4-ACA interactions much weaker than Me-4PACz self-interactions. As a result, parallel and angular Me/Me interactions remain dominant. Comparatively, in the Me/4-APA and Me/4-ASA systems, parallel interactions are the strongest, suggesting that 4-APA and 4-ASA co-assemble favorably with Me-4PACz into ordered co-SAMs. These interactions are also stronger than Me/Me self-interactions, indicating that 4-APA and 4-ASA can disrupt Me-4PACz micellization and relieve molecular aggregation, consistent with dynamic light scattering (DLS) measurements (Supplementary Fig. 9). However, in the Me/4-APA system, steric hindrance originating from  $-\text{PO}_3\text{H}_2$  groups induces strong inter-headgroup repulsion, which suppresses linear intermolecular interactions (−1.14 eV). Consequently, angular 4-APA/4-APA (−1.21 eV) and Me-Me (−1.19 eV) interactions remain important and still promote self-aggregation. By contrast, in the Me/4-ASA system, the  $-\text{SO}_3\text{H}$  groups demonstrate highly delocalized negative charges, minimizing electrostatic repulsion between adjacent headgroups. Their tetrahedral geometry further enables stable parallel (−1.37 eV) and linear (−1.24 eV) pairing with Me-4PACz, facilitating more ordered co-SAMs formation.

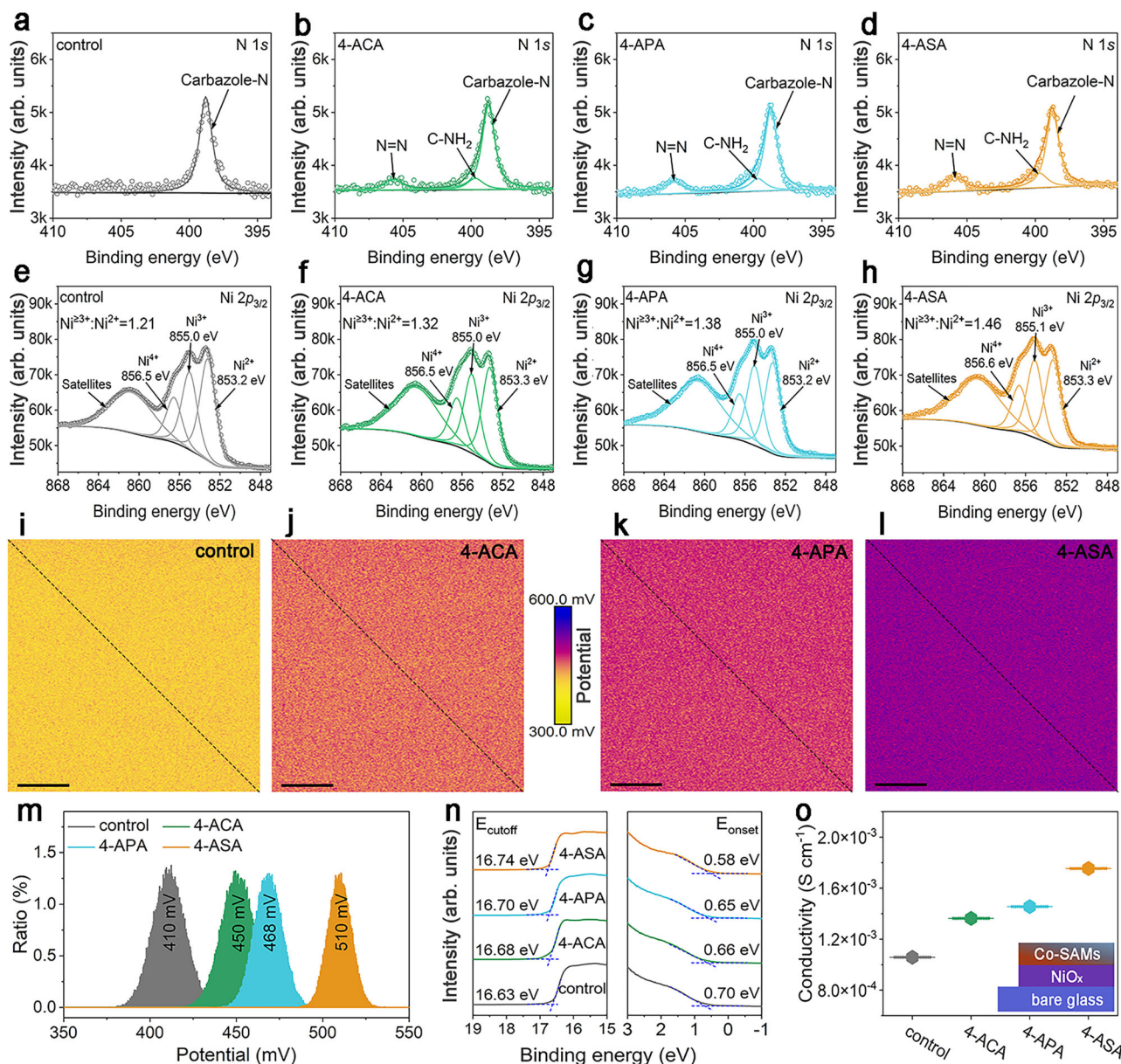
### Effects of co-SAMs on the Properties of $\text{NiO}_x$

To investigate co-SAMs effects on  $\text{NiO}_x$ , X-ray photoelectron spectroscopy (XPS) measurements were conducted to analyze chemical states of the modified substrates. The control sample exhibits characteristic C–C and C–P peaks, confirming the presence of Me-4PACz. In the co-assembled monolayers, 4-ACA was evidenced by the O–C=O signal, 4-APA by an intensified C–P peak, and 4-ASA was identified from both

C 1s and S 2p spectra (Supplementary Fig. 10). Analysis of the N 1s spectra (Fig. 2a–d) further revealed two additional nitrogen species in the co-SAMs, assigned to  $-\text{NH}_2$  and azo ( $\text{N}=\text{N}$ ) groups derived from the co-adsorbents<sup>26</sup>. The atomic ratio  $\text{N}_{(\text{amine} + \text{N}=\text{N})}/\text{N}_{(\text{carbazole})}$  quantifies co-adsorbent incorporation (Supplementary Fig. 11), showing the highest content for 4-ASA and the lowest for 4-ACA. Besides, Ni 2p XPS spectra were systematically analyzed. As shown in Fig. 2e–h, the Ni  $2p_{3/2}$  core spectra were deconvoluted into three peaks at approximately 856.5, 855, and 853.2 eV, corresponding to  $\text{Ni}^{4+}$ ,  $\text{Ni}^{3+}$ , and  $\text{Ni}^{2+}$ , respectively<sup>27</sup>. Semi-quantitative analysis showed that the  $\text{Ni}^{3+}/\text{Ni}^{2+}$  ratio increased progressively from 1.21 in the control to 1.32, 1.38 and 1.46 for 4-ACA, 4-APA and 4-ASA, respectively. This increase indicates a shift in the Ni oxidation-state distribution toward higher-valent species, which may contribute to enhanced hole-transport properties of the composite HTL<sup>28</sup>.

Subsequently, the surface morphology and optoelectronic properties of the modified  $\text{NiO}_x$  substrates were investigated. Scanning electron microscopy (SEM) images indicate that co-SAMs modification does not alter the morphology of the  $\text{NiO}_x$  films (Supplementary Fig. 12). In contrast, atomic force microscopy (AFM) revealed reduced surface roughness, with root-mean-square (RMS) values decreasing from 5.1 nm for the control film to 5.0 nm for 4-ACA, 4.5 nm for 4-APA and 3.9 nm for 4-ASA. The lowest roughness for the 4-ASA film further suggests that 4-ASA-based co-SAMs enable improved coverage of the  $\text{NiO}_x$  surface and suppress SAM self-segregation. This morphological optimization is a key metric for efficient charge extraction in PSCs<sup>29</sup>. Kelvin probe force microscopy (KPFM) under compensated bias was further used to evaluate the surface potential distribution of the films (Fig. 2i–l). Line profile analysis of contact potential difference (CPD) (Supplementary Fig. 13) shows reduced variation from 65 mV (control) to 55 mV (4-ACA), 50 mV (4-APA), and 30 mV (4-ASA), indicating a more homogeneous potential distribution after co-SAMs modification. This homogeneity primarily originates from surface defect passivation via competitive coordination. Co-SAMs modification also increased the mean CPD from 410 mV for the control film to 450 mV for 4-ACA, 468 mV for 4-APA and 510 mV for 4-ASA (Fig. 2m). According to the relationship  $\text{CPD} = (\phi_{\text{tip}} - \phi_{\text{sample}})/e^{30,31}$ , where  $\phi$  denotes work function, the increased CPD corresponds to a reduced  $\text{NiO}_x$  work function ( $\phi_{\text{sample}}$ ). Ultraviolet photoelectron spectroscopy (UPS) was used to quantify the work function of the films. Figure 2n displays the secondary electron cutoff ( $E_{\text{cutoff}}$ ) and valence band onset ( $E_{\text{onset}}$ ) energies for the control, 4-ACA, 4-APA, and 4-ASA films. The corresponding Fermi levels ( $E_F$ ) were −4.59, −4.54, −4.52, and −4.48 eV, respectively (Supplementary Table 1). The upward shift in  $E_F$  for the co-SAMs modified films indicates a reduced work function, corroborating the KPFM results. Notably,  $E_{\text{onset}}$  corresponds to the energy difference between the  $E_{\text{VBM}}$  and  $E_F$ . The smallest  $E_{\text{onset}}$  value of 0.58 eV observed for the 4-ASA film suggests the most pronounced p-type doping, consistent with its highest  $\text{Ni}^{3+}/\text{Ni}^{2+}$  ratio (1.46) confirmed by XPS analysis. Although previous studies have associated higher  $\text{Ni}^{3+}/\text{Ni}^{2+}$  ratios with increased work function and a deeper  $E_{\text{VBM}}$  in  $\text{NiO}_x$ <sup>21,27</sup>, our co-SAMs-modified systems exhibit a contrasting trend (Supplementary Fig. 14). This behavior is primarily attributed to interfacial dipoles induced by 4-ACA, 4-APA, and 4-ASA, which lower the vacuum level. The resulting reduction in work function and elevation of  $E_{\text{VBM}}$  critically minimize the energy offset at the  $\text{NiO}_x/\text{Pb-Sn}$  perovskite interface, thereby facilitating hole extraction in photovoltaic devices.

The conductivity of the modified  $\text{NiO}_x$  substrates was systematically evaluated by measuring the resistivity of bare glass/ $\text{NiO}_x$ /co-SAMs films at room temperature using a Physical Property Measurement System (PPMS-9) (Fig. 2o). The results suggest that co-SAMs modification significantly increased the conductivity of  $\text{NiO}_x$  films from  $1.06 \times 10^{-3} \text{ S cm}^{-1}$  for the control film to  $1.36 \times 10^{-3} \text{ S cm}^{-1}$  for 4-ACA,  $1.45 \times 10^{-3} \text{ S cm}^{-1}$  for 4-APA, and  $1.75 \times 10^{-3} \text{ S cm}^{-1}$  for 4-ASA. Complementary current-voltage (I–V) measurements on ITO/ $\text{NiO}_x$ /co-



**Fig. 2 | Effect of co-SAMs on  $\text{NiO}_x$ .** **a–d** XPS N 1s spectra and **e–h** Ni 2p<sub>3/2</sub> spectra for control, 4-ACA, 4-APA, and 4-ASA films, respectively. **i–l** The surface KPFM results of the modified  $\text{NiO}_x$  substrates. The scale bar is 2  $\mu\text{m}$ . **m** Potential distribution plots of

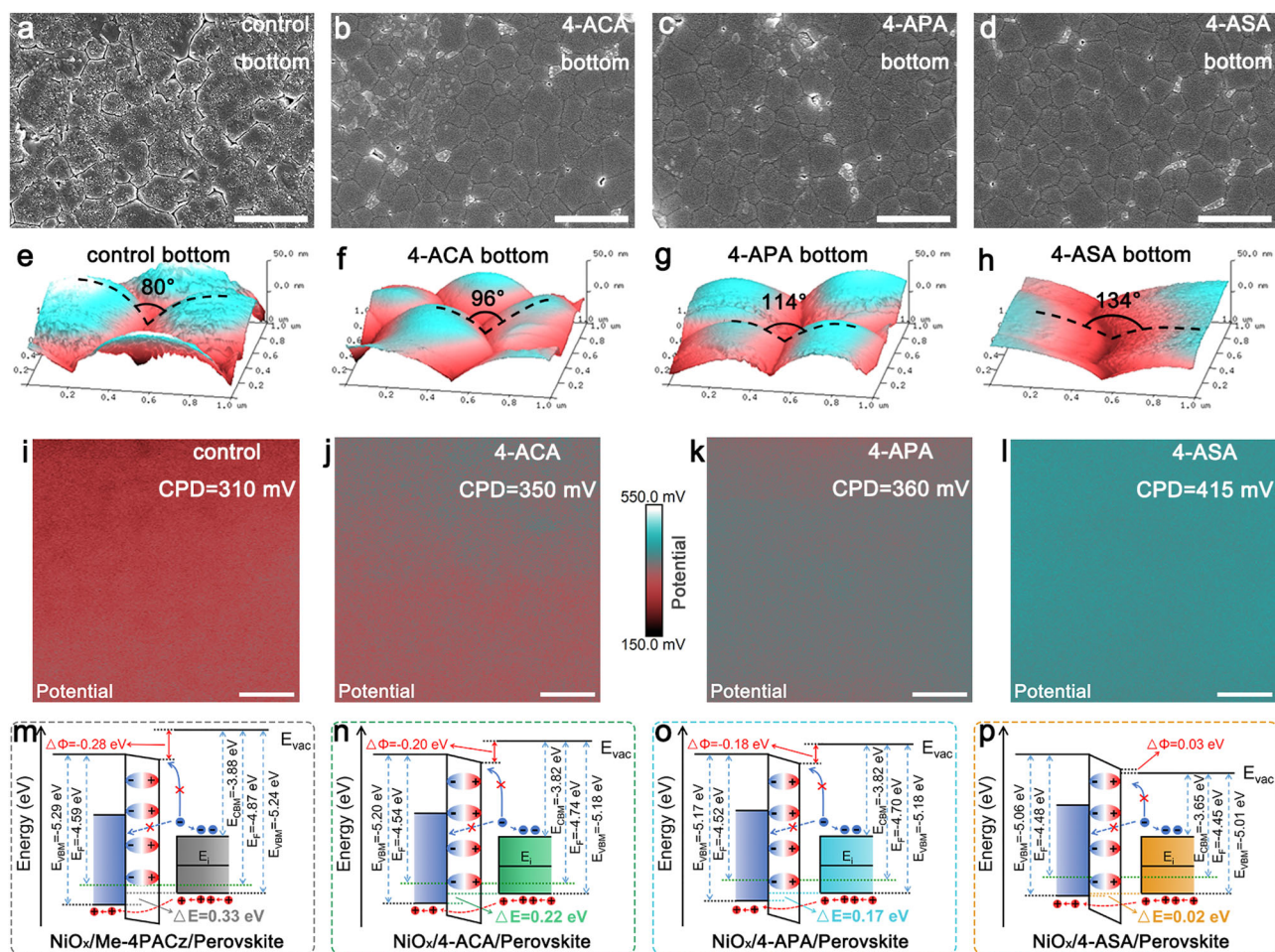
control, 4-ACA, 4-APA, and 4-ASA films. **n** UPS spectra of control, 4-ACA, 4-APA, and 4-ASA films. **o** Conductivity measurements via PPMS-9 system at room temperature for control, 4-ACA, 4-APA, and 4-ASA films, respectively.

SAMs/Au effective stacks further confirm the conductivity enhancement (Supplementary Fig. 15), in agreement with the PPMS results. These conductivity measurements suggest that co-SAMs modification improves the conductivity of  $\text{NiO}_x/\text{co-SAMs}$  films, and thus charge-transport properties, consistent with the increased  $\text{Ni}^{3+}/\text{Ni}^{2+}$  ratio revealed by XPS. Besides, despite the presence of light-absorbing azo chromophores, with the corresponding solutions showing a light-yellow color (Supplementary Fig. 16), the co-SAMs-modified films retained average transmittance comparable to that of the control across the 400–900 nm range. This exceptional optical transparency, arising from the low concentration (0.3 mg mL<sup>-1</sup>) and sub-monolayer molecular coverage, is favorable for high-efficiency perovskite solar cells. The surface wettability of the modified substrates was further assessed by static contact angle measurements using anhydrous DMF (Supplementary Fig. 17). The contact angle decreased from 47.5° (control) to 39.0° (4-ACA), 27.5° (4-APA), and 17.5° (4-ASA), indicating enhanced hydrophilicity after co-SAMs modification due to the

exposure of hydrophilic –NH<sub>2</sub> groups. This improved wettability promotes uniform spreading of the perovskite precursor during spin-coating.

### Evaluation of perovskite thin films

It is well established that the uniformity, chemical environment, and wettability of the underlying substrate critically influence the crystallinity and morphology of perovskite films. To examine the effect of co-SAMs modification on Pb-Sn-based inorganic perovskite crystallization, the perovskite films were exfoliated from their substrates. Figure 3a–d presents the bottom-view SEM images of the exfoliated films. Notably, the control film exhibits a rough bottom surface with abundant nanoparticles, which were identified as  $\text{NiO}_x$  by energy-dispersive X-ray (EDX) elemental mapping (Supplementary Fig. 18). Their presence indicates that pure Me-4PACz modification results in incomplete coverage of the  $\text{NiO}_x$  surface, leading to direct contact between  $\text{NiO}_x$  and the Pb-Sn perovskite. Interestingly, these  $\text{NiO}_x$



**Fig. 3 | Effect of co-SAMs on perovskite.** **a–d** Bottom-view SEM images of the control, 4-ACA, 4-APA, and 4-ASA perovskite films, respectively. The scale bar is 1  $\mu\text{m}$ . **e–h** High-resolution AFM images of the buried interfaces for different perovskite films. **i–l** The surface KPFM results of perovskite films deposited on

different substrates. The scale bar is 1  $\mu\text{m}$ . **m–p** Schematic energy level diagrams of (**m**)  $\text{NiO}_x/\text{Me-4PACz}/\text{perovskite}$ , (**n**)  $\text{NiO}_x/4\text{-ACA}/\text{perovskite}$  (**o**)  $\text{NiO}_x/4\text{-APA}/\text{perovskite}$ , and (**p**)  $\text{NiO}_x/4\text{-ASA}/\text{perovskite}$  interfaces.

nanoparticles are substantially reduced in the 4-ACA and 4-APA perovskite films and entirely absent in the 4-ASA perovskite film, indicating that co-SAMs modification, particularly with 4-ASA, affords improved  $\text{NiO}_x$  surface coverage, which aligns well with the MD simulation results (Fig. 1c). The cross-sectional morphology of the perovskite films was also examined (Supplementary Fig. 19). It can be seen that co-SAMs modification, especially with 4-ASA, significantly improves the compactness and crystallinity of the perovskite layer, with fewer grain boundaries and reduced voids compared to the control film. To gain deeper insights into the impact of co-SAMs modification at the buried interface, perovskite grain boundary groove (GBG) angles were analyzed, which form during residual-solvent evaporation and direct top-down grain growth<sup>32,33</sup>. Figure 3e–h displays height profiles of exfoliated buried interfaces, acquired via high-resolution AFM. The control film exhibits a GBG angle of 80°, whereas introduction of the tailored co-adsorbents progressively increased the angle to 96° for 4-ACA, 114° for 4-APA, and 134° for 4-ASA. The increased GBG angles indicate enhanced heterointerface energy<sup>34</sup>, which is primarily attributed to improved surface hydrophilicity and specific interactions between  $-\text{NH}_2$  groups of the tailored molecules and perovskite precursors, as supported by DFT calculations (Supplementary Fig. 20), Fourier-transform infrared (FTIR) spectroscopy (Supplementary Fig. 21), and nuclear magnetic resonance (NMR) spectroscopy (Supplementary Fig. 22)<sup>35–37</sup>. This significant increase in GBG angles is critical for inhibiting nanovoid formation at the buried

interfaces, thereby reducing trap density and non-radiative recombination centers.

Further, the chemical states at the bottom surface of the exfoliated perovskite films were investigated (Supplementary Fig. 23). The control perovskite film exhibits distinct  $\text{Ni}2p_{3/2}$  XPS peaks, confirming that the nanoparticles observed in SEM images correspond to  $\text{NiO}_x$ . Upon co-SAMs modification, the  $\text{Ni}2p_{3/2}$  peak intensity exhibits significant attenuation, providing additional evidence of improved  $\text{NiO}_x$  surface coverage. Besides, relative to the control, the co-SAMs-modified perovskites demonstrate an upshift in the  $\text{I}3d$  binding energy together with corresponding downshifts in the  $\text{Pb}4f$  and  $\text{Sn}3d$  signals. These reciprocal shifts can be attributed to the altered interfacial coordination and the reconstructed interfacial electronic structure triggered by co-SAMs modification, which may promote electron redistribution between the  $-\text{NH}_2$  groups and  $\text{Pb}/\text{Sn}$  sites<sup>38</sup>. Such electronic interactions suggest enhanced co-SAMs/perovskite coordination, potentially stabilizing  $\text{Sn}^{2+}$  species at the buried interface<sup>39</sup>. To verify this, the  $\text{Sn}3d$  core spectra were deconvoluted into two components assigned to  $\text{Sn}^{2+}$  (black peaks) and  $\text{Sn}^{4+}$  (red peaks). The  $\text{Sn}^{4+}$  content, derived from the perspective peak areas, decreased significantly in co-SAMs-modified perovskites compared to the control. Collectively, XPS results verify that co-SAMs modification effectively suppresses  $\text{Sn}^{2+}$  oxidation at the buried interface through two synergistic mechanisms. First, improved surface coverage of  $\text{NiO}_x$  isolates it from the  $\text{Pb-Sn}$  perovskite, inhibiting redox reactions between  $\text{Ni}^{2+3+}$

and  $\text{Sn}^{2+}$ . Second, co-SAMs modification triggered interfacial coordination changes that change the chemical environment around  $\text{Sn}^{2+}$ , thermodynamically suppressing its oxidation to  $\text{Sn}^{4+}$ . Additionally, residual stress of the perovskite films was investigated using grazing incidence X-ray diffraction (GIXRD) (Supplementary Fig. 24)<sup>40</sup>. At a fixed grazing incidence angle of  $1.5^\circ$  with  $\psi$  varying from  $10^\circ$  to  $50^\circ$ , the control perovskite exhibits systematic peak shifts toward lower angles, indicating significant tensile stress. In contrast, perovskites on co-SAMs-modified substrates demonstrate progressively attenuated peak shifts with increasing  $\psi$ . This tensile stress reduction is attributed to passivation of iodine vacancies ( $\text{V}_\text{I}$ ) at the buried interface by  $-\text{NH}_2$  groups of tailored co-adsorbent molecules, as confirmed by DFT calculations (Supplementary Fig. 25). The substantial stress reduction in co-SAM-modified perovskites enhances photovoltaic performance and long-term stability of PSCs, as residual stress primarily contributes to film cracking and interfacial degradation<sup>41</sup>.

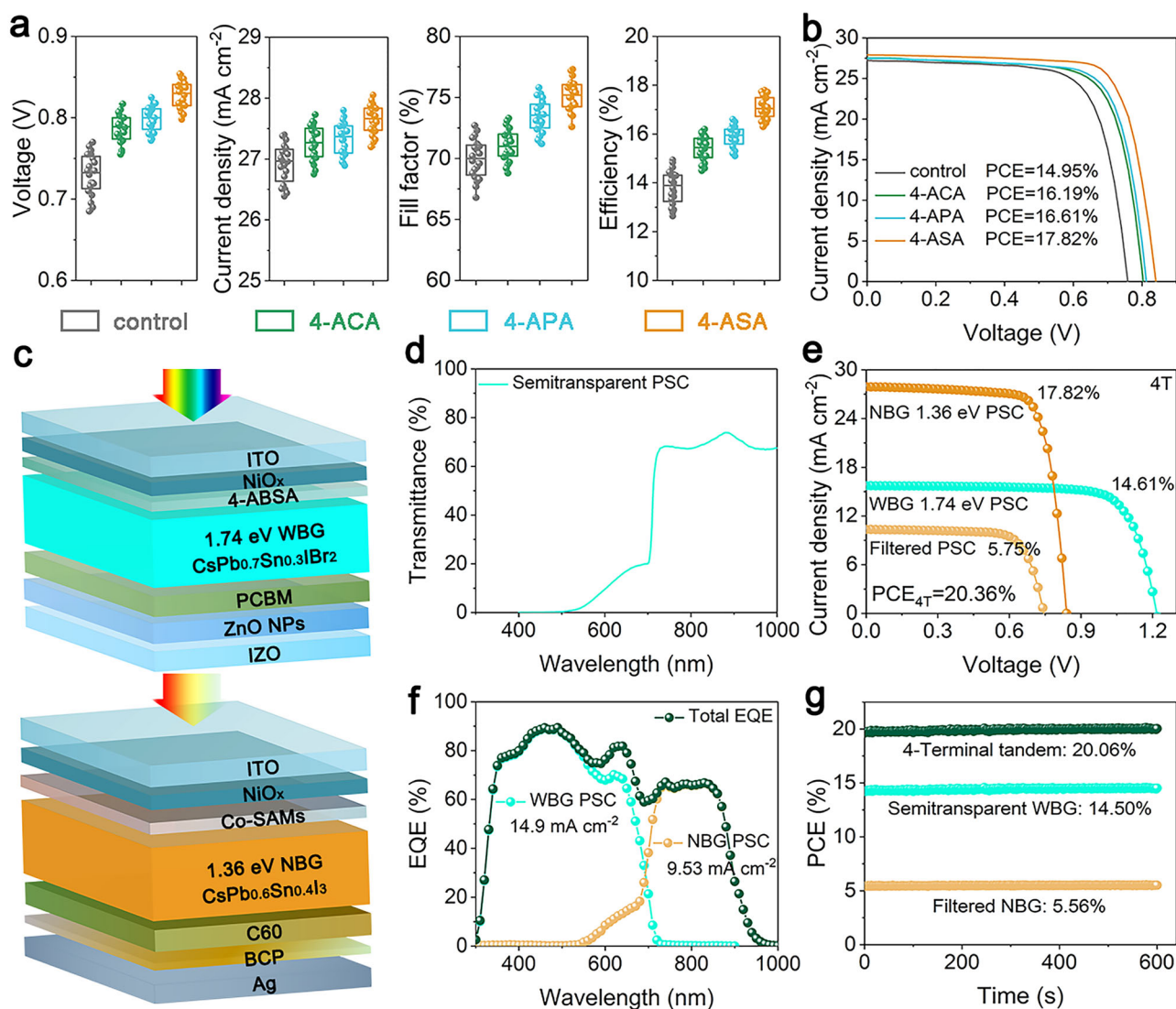
Top-surface morphologies of the perovskite films were characterized by top-view SEM (Supplementary Fig. 26). All films exhibit full substrate coverage, indicating uniform perovskite deposition. Statistical grain-size analysis reveals a marked increase in average grain size (AGS) for films deposited on co-SAM-modified substrates. This grain enlargement is primarily attributed to Lewis acid-base interactions between the  $-\text{NH}_2$  groups of tailored co-adsorbents and the perovskite precursors. These interactions passivate vacancy defects while concurrently modulating crystallization kinetics of Pb- and Sn-based components, thereby promoting the formation of larger, more uniform perovskite grains with enhanced crystallinity, as supported by the XRD and UV results (Supplementary Fig. 27)<sup>42</sup>. AFM measurements further complement the SEM analysis, revealing a progressive decrease in RMS roughness from 24.0 nm (control) to 23.4 nm (4-ACA), 16.1 nm (4-APA), and 12.9 nm (4-ASA) (Supplementary Fig. 28). The 4-ASA perovskite film displays the lowest surface roughness, which is beneficial for minimizing leakage currents and suppressing interfacial non-radiative recombination<sup>43</sup>. Figure 3i–l presents the corresponding KPFM results, with CPD increasing from 310 mV for the control film to 350 mV for 4-ACA, 360 mV for 4-APA, and 415 mV for 4-ASA. As established earlier, a higher CPD corresponds to a lower work function. UPS was further used to determine the energy levels of the different perovskite films (Supplementary Fig. 29 and Table 2). The results confirm a progressive reduction in work function from 4.87 eV (control) to 4.74 eV (4-ACA), 4.70 eV (4-APA), and 4.45 eV (4-ASA), consistent with the KPFM observations. Besides, the energy gap between  $E_{\text{VBM}}$  and  $E_\text{F}$  for the 4-ACA (0.44 eV), 4-APA (0.48 eV), and 4-ASA (0.56 eV) perovskite films is significantly larger than that of the control film (0.37 eV), indicating a de-doping process due to the inhibition of interfacial  $\text{Sn}^{2+}$  oxidation as discussed earlier<sup>44</sup>. Figure 3m–p illustrates the energy level diagrams of the  $\text{NiO}_x$ /perovskite interfaces modified with pure Me-4PACz versus co-SAMs. At the  $\text{NiO}_x$ /Me-4PACz/perovskite interface, the hole-transport energy-level offset ( $\Delta E$ ) is 0.33 eV, indicating a substantial barrier. Upon co-SAMs modification,  $\Delta E$  progressively decreases to 0.22 eV ( $\text{NiO}_x$ /4-ACA/perovskite), 0.17 eV ( $\text{NiO}_x$ /4-APA/perovskite), and ultimately 0.02 eV ( $\text{NiO}_x$ /4-ASA/perovskite). This systematic reduction mainly arises from the dipole-engineered co-SAM layers, which lower the work function of  $\text{NiO}_x$ , as discussed above. The minimal  $\Delta E$  (0.02 eV) achieved for  $\text{NiO}_x$ /4-ASA/perovskite affords near-ideal energy level alignment for hole extraction, which is essential for high-performance PSCs.

### Performance of Pb-Sn-based inorganic PSCs

To evaluate the effect of co-SAMs on device performance, inverted NBG Pb-Sn-based inorganic PSCs featuring  $\text{CsPb}_{0.6}\text{Sn}_{0.4}\text{I}_3$  active layers and ITO/ $\text{NiO}_x$ /co-SAMs/perovskite/C60/BCP/Ag architecture were fabricated. Me-4PACz was fixed at  $0.5 \text{ mg mL}^{-1}$  as the control, whereas 4-ACA, 4-APA, and 4-ASA were optimized to  $0.3 \text{ mg mL}^{-1}$ , at which all devices showed the highest efficiencies (Supplementary Fig. 30), likely

owing to insufficient coverage at low concentration and aggregation-induced defects at high concentration. Figure 4a summarizes statistical photovoltaic parameters from 20 devices fabricated in four independent batches (5 devices per batch). All co-SAMs-modified devices exhibited consistently improved open-circuit voltage ( $V_{\text{OC}}$ ), short-circuit current density ( $J_{\text{SC}}$ ), fill factor (FF), and PCE relative to the control, with 4-ASA performing best. The champion 4-ASA device delivered a scanned PCE of 17.82%, with  $V_{\text{OC}} = 0.84 \text{ V}$  and  $J_{\text{SC}} = 27.9 \text{ mA cm}^{-2}$  (Fig. 4b and Supplementary Table 3). Such a PCE represents the highest scanned efficiency recorded for PSCs utilizing Cs-based Pb-Sn-mixed inorganic perovskite absorber thus far (Supplementary Fig. 31). This substantial performance enhancement primarily originates from the minimal energy level offset ( $\Delta E = 0.02 \text{ eV}$ ) enabled by the strong dipole moment of 4-ASA (8.8 D) at the  $\text{NiO}_x$ /4-ASA/perovskite interface, which facilitates efficient charge carrier extraction and suppresses non-radiative recombination. To quantitatively evaluate non-radiative recombination, photoluminescence (PL), time-resolved photoluminescence (TRPL), and photoluminescence quantum yield (PLQY) measurements were conducted on  $\text{NiO}_x$ /co-SAMs/perovskite structures (Supplementary Fig. 32). Among all samples, the 4-ASA-based perovskite shows the strongest PL quenching, the shortest carrier lifetime (6.9 ns), and the highest PLQY (0.71%), implying more efficient interfacial charge extraction together with suppressed non-radiative recombination<sup>45</sup>. Besides, external quantum efficiency (EQE) and steady-state power output (SPO) results (Supplementary Fig. 33) are consistent with the J-V curves, confirming negligible overestimation of photovoltaic parameters. Hysteresis behavior was quantified via the hysteresis index (HI) (Supplementary Fig. 34)<sup>46</sup>. The HI values were 10.5% (control), 7.6% (4-ACA), 4.3% (4-APA), and 4.1% (4-ASA), showing progressive hysteresis reduction with co-SAMs modification. The minimal hysteresis of the 4-ASA device is attributed to its improved interfacial properties, including passivated defects, enlarged GBG angles, and optimized energy-level alignment.

To further improve the efficiency of Pb-Sn-based inorganic PSCs, NBG PSCs were integrated into all-perovskite 4 T TSCs. The TSCs were fabricated by mechanically stacking semi-transparent 1.74 eV WBG  $\text{CsPb}_{0.7}\text{Sn}_{0.3}\text{IBr}_2$  top cells and 1.36 eV NBG  $\text{CsPb}_{0.6}\text{Sn}_{0.4}\text{I}_3$  bottom cells (Fig. 4c). Notably, although the 4-ASA co-SAMs system delivers better performance than 4-ABSA in the NBG subcell due to the somewhat improved hole extraction and surface coverage, 4-ABSA remains more suitable for the WBG subcell because of its better compatibility with the corresponding perovskite (Supplementary Fig. 35). According to the Shockley-Queisser (SQ) limit, the optimal bandgap combination for 4T TSCs is approximately 1.30 eV (NBG) and 1.80 eV (WBG), which optimizes spectral utilization and reduces photon loss. Nevertheless, this ideal pairing cannot be applied to Cs-based inorganic Pb-Sn perovskite absorbers due to inherent material limitations. The choice of 1.36 eV NBG subcell ( $\text{CsPb}_{0.6}\text{Sn}_{0.4}\text{I}_3$ ) is imposed by the fundamental instability of narrower band gaps in such systems, where achieving NBG < 1.36 eV requires Sn content exceeding 40%. Such elevated Sn concentrations accelerate severe  $\text{Sn}^{2+}$  oxidation and  $\text{CsSnI}_3/\text{CsPbI}_3$  phase segregation, adversely degrading device performance. For the WBG subcell,  $\text{CsPb}_{0.7}\text{Sn}_{0.3}\text{IBr}_2$  (1.74 eV) was chosen to balance lead reduction, phase stability, and bandgap demands. A bandgap above 1.74 eV would require more lead or bromine. Excessive bromine causes severe lattice strain, driving the transition from photoactive cubic phase to inactive orthorhombic phase. XRD, SEM, UV-vis, and Tauc's plot results (Supplementary Fig. 36) demonstrate that the 1.74 eV WBG  $\text{CsPb}_{0.7}\text{Sn}_{0.3}\text{IBr}_2$  perovskite film exhibits excellent crystallinity. The transmittance of the semi-transparent WBG PSCs reached approximately 70% over the 730–1000 nm wavelength range (Fig. 4d) without significant spectral distortion, enabling efficient photon harvesting by the NBG bottom subcell. Under reverse scan, the semi-transparent WBG PSCs achieve a champion PCE of 14.61%, with a  $V_{\text{OC}}$



**Fig. 4 | Photovoltaic performance of 1.36 eV NBG Pb-Sn-based inorganic PSCs and 4 T TSCs.** **a** Statistical  $V_{OC}$ ,  $J_{SC}$ , FF, and PCE of NBG Pb-Sn-based inorganic PSCs. **b** J-V curves of champion control, 4-ACA, 4-APA, and 4-ASA-based devices under reverse scan. **c** Device structure of the semi-transparent 1.74 eV WBG and 1.36 eV NBG cells for 4T TSCs. **d** UV-vis transmittance spectra of semi-transparent WBG

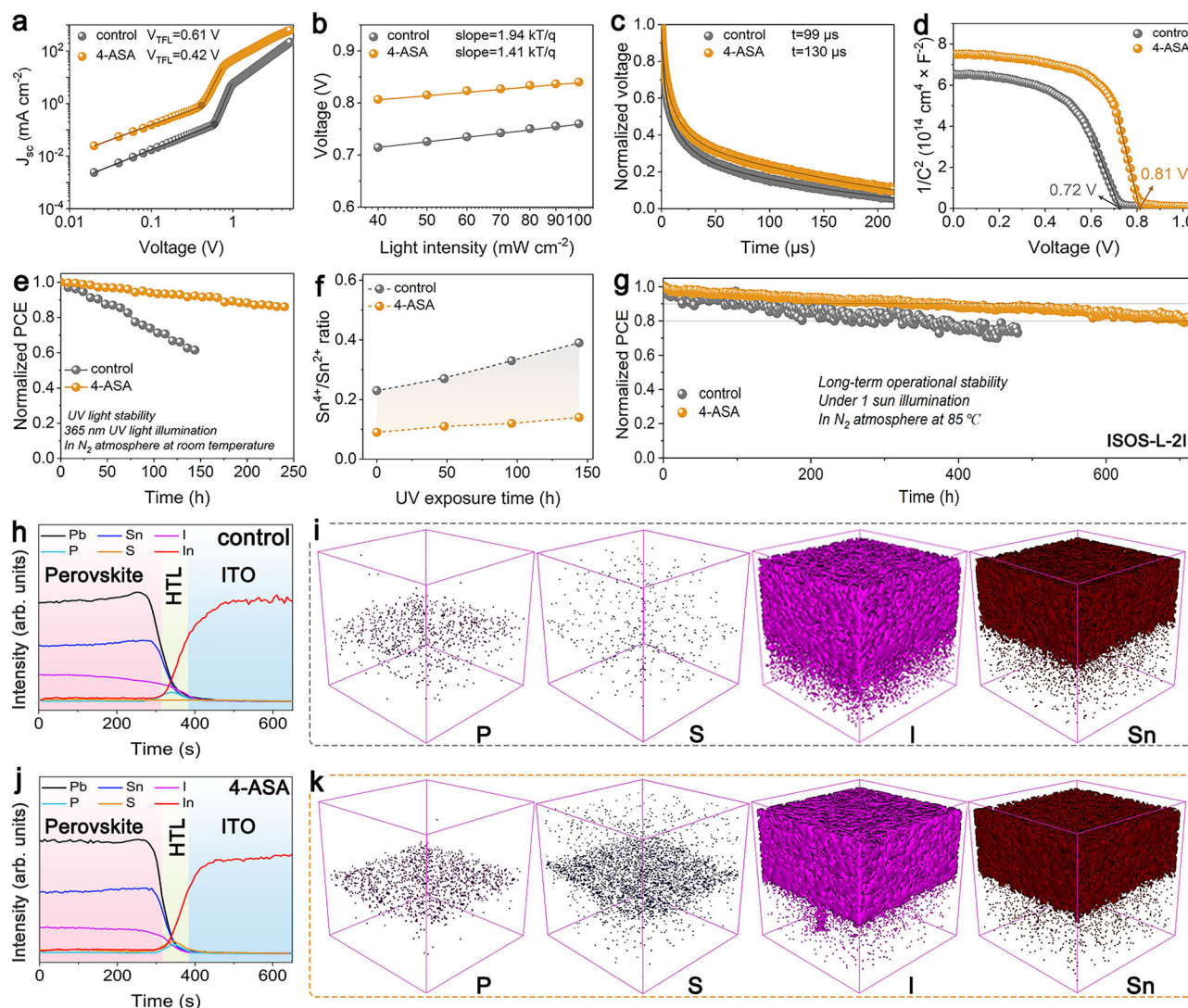
$CsPb_{0.7}Sn_{0.3}I_{Br_2}$  PSCs. **e** J-V curves of the 4T TSCs with semi-transparent WBG PSCs, NBG PSCs and filtered NBG PSCs. **f** EQE spectra of 4 T TSCs with semi-transparent WBG PSCs and filtered NBG PSCs. **g** Steady-state PCE of the 4T TSCs with semi-transparent WBG PSCs and filtered NBG PSCs.

of 1.21 V,  $J_{SC}$  of  $15.74 \text{ mA cm}^{-2}$ , and FF of 76.7%. Upon stacking the semi-transparent WBG device on the NBG bottom cell, the filtered NBG bottom cell achieves a PCE of 5.75% ( $V_{OC}=0.75 \text{ V}$ ,  $J_{SC}=10.35 \text{ mA cm}^{-2}$ , and FF=74.1%). The resultant 4 T all-perovskite TSCs achieves an overall scanned PCE of 20.36% (Fig. 4e). The EQE spectra indicate integrated  $J_{SC}$  values of  $14.9 \text{ mA cm}^{-2}$  and  $9.53 \text{ mA cm}^{-2}$  for the semi-transparent WBG PSCs and filtered NBG bottom cell, respectively (Fig. 4f). Under constant AM 1.5 G illumination at MPP for 600 s, the semi-transparent WBG PSCs and filtered NBG device exhibit SPO of 14.50% and 5.56%, respectively, while the 4 T all-perovskite TSC achieved an SPO of 20.06% (Fig. 4g). This configuration represents, to the best of our knowledge, the first 4T Cs-based inorganic perovskite TSCs where both the top (WBG  $CsPb_{0.7}Sn_{0.3}I_{Br_2}$ ) and bottom (NBG  $CsPb_{0.6}Sn_{0.4}I_3$ ) subcells employ Pb-Sn mixed perovskite absorbers. Prior 4T all-perovskite tandems have typically paired Pb-only WBG top cells with Pb-Sn NBG bottom cells, or employed hybrid organic-inorganic perovskite subcells, Pb-Sn/Pb-Sn 4 T tandems assembled with Cs-based inorganic perovskite absorbers have not been previously demonstrated<sup>47–49</sup>. Besides,

scaled-up devices with  $1 \text{ cm}^2$  active area were further fabricated (Supplementary Fig. 37), and the corresponding NBG Pb-Sn mixed perovskite sub-cell delivers an efficiency of 15.3%, while the WBG device achieves 13.1%. Following integration, the 4 T TSCs ( $1 \text{ cm}^2$ ) yield a champion efficiency of 17.6%. These findings confirm that our co-SAMs modification strategy maintains high performance upon small-scale amplification, establishing a foundation for future industrial-compatible large-area fabrication.

#### Charge carrier dynamics and stability of PSCs

To gain deeper insights into the significant efficiency enhancement, charge carrier dynamics were investigated with a primary focus on defect density, as trap states directly promote recombination and impede carrier transport. Space-charge-limited current (SCLC) technique was employed to quantitatively estimate defect densities ( $N_t$ ) in perovskite films<sup>50</sup>. Figure 5a displays the dark J-V curves of hole-only devices, where the 4-ASA-modified device exhibits a smaller trap-filled limit voltage ( $V_{TFL}$ ) of 0.42 V compared to 0.61 V for the control device, yielding a decreased  $N_t$  from  $8.42 \times 10^{15} \text{ cm}^{-3}$  to  $5.79 \times 10^{15} \text{ cm}^{-3}$  after



**Fig. 5 | Charge dynamics and stability of control and 4-ASA devices.** **a** Dark J-V curves of hole-only devices based on different perovskite films. **b**  $V_{OC}$  response under varying light intensities, **(c)** TPV results, and **(d)** Mott-Schottky plots for control and 4-ASA devices. **e** UV light stability of the devices measured under 365 nm UV light exposure at room temperature. **f** Statistics of the  $Sn^{4+}/Sn^{2+}$  ratio

extracted from XPS spectra after different UV light exposure durations. **g** Operational stability of unencapsulated control and 4-ASA devices under 1 sun illumination in N<sub>2</sub> atmosphere at 85 °C. **h, j** TOF-SIMS depth profiles of the perovskite/Me-4PACz (with or without 4-ASA)/NiO<sub>x</sub>/ITO. **i, k** 3D images for P, S, I, and Sn elemental distribution in control and 4-ASA samples.

4-ASA treatment (Supplementary Table 4). This substantial reduction in defect density originates primarily from suppressed  $Sn^{2+}$  oxidation, passivated  $V_i$  defects at the NiO<sub>x</sub>/perovskite interface, and improved perovskite crystallinity. Complementing the defect density analysis, charge carrier recombination and extraction kinetics were evaluated by light-intensity-dependent photovoltaic measurements. As shown in Fig. 5b, the 4-ASA device exhibits a smaller  $V_{OC}$  versus light-intensity slope (1.41 kT/q) than the control (1.94 kT/q), indicating effective suppression of trap-assisted recombination<sup>51</sup>. Meanwhile, the dependence of  $J_{SC}$  on light intensity (Supplementary Fig. 38) yields a near-unity  $\alpha$  value of 0.955 for the 4-ASA device, signifying enhanced charge extraction efficiency<sup>52</sup>. Besides, transient photovoltage (TPV) and transient photocurrent (TPC) measurements were conducted to further quantify charge recombination and extraction<sup>53</sup>. The TPV results (Fig. 5c) demonstrate that the control device exhibits a faster recombination decay (99 μs vs. 130 μs for 4-ASA), reflecting more pronounced recombination resulting from suboptimal NiO<sub>x</sub>/perovskite interfacial properties. The TPC results (Supplementary Fig. 39) reveal accelerated charge extraction from 0.91 μs to 0.55 μs due to the enhanced conductivity of the modified interface and optimized energy

level alignment following 4-ASA modification, correlating with the improved  $J_{SC}$  in 4-ASA devices. Additionally, Mott-Schottky analysis was conducted on capacitance-voltage (C-V) profiles to determine the built-in potential ( $V_{bi}$ )<sup>54</sup>. As shown in Fig. 5d, the  $V_{bi}$  increased from 0.72 V in the control device to 0.81 V in the 4-ASA device, directly contributing to the observed  $V_{OC}$  improvement. Collectively, the remarkable performance enhancement of the 4-ASA devices can be attributed to suppressed charge recombination and enhanced charge extraction, which are enabled by reduced trap-state density and improved interfacial energetics.

Beyond device efficiency, stability constitutes another critical metric for evaluating photovoltaic cell performance. To examine the beneficial effects of co-SAMs modification on device stability, the PCE of unencapsulated NBG devices was tracked under 365 nm ultraviolet (UV) light irradiation in a nitrogen (N<sub>2</sub>) atmosphere at room temperature. As shown in Fig. 5e, the control device degrades rapidly to 61.6% of initial PCE after 144 hours, whereas the 4-ASA device maintains 86.1% of initial efficiency after 240 hours, demonstrating significantly enhanced UV stability. This improvement is primarily attributed to two factors. First, co-SAMs assembly can significantly

suppress C-N bond cleavage and structural decomposition of Me-4PACz under UV irradiation via intermolecular hydrogen bonding (Supplementary Fig. 40), thereby inhibiting the generation of interfacial defects caused by SAM decomposition. Second, improved surface coverage of the  $\text{NiO}_x$  substrate by 4-ASA-based co-SAMs forms a protective physical barrier that inhibits direct  $\text{NiO}_x/\text{Pb-Sn}$  perovskite contact, thereby suppressing  $\text{Sn}^{2+}$  oxidation induced by under-coordinated  $\text{Ni}^{2+}$  species under UV illumination<sup>55</sup>. To verify this, the chemical states of the  $\text{NiO}_x$ /perovskite interface were investigated under varying durations of UV exposure. Figure 5f shows the time-dependent  $\text{Sn}^{4+}/\text{Sn}^{2+}$  ratios at the bottom surface of control and 4-ASA perovskite films, where ratio values were extracted from XPS measurements (Supplementary Fig. 41). Following 144 hours of UV irradiation, the control film exhibits a substantial increase in  $\text{Sn}^{4+}/\text{Sn}^{2+}$  ratio from 0.23 to 0.39 (a 69.5% increase relative to initial values), confirming UV-accelerated redox reactions between  $\text{Ni}^{2+}$  and  $\text{Sn}^{2+}$ . In contrast, the 4-ASA perovskite film maintains greater stability with a ratio change from 0.09 to 0.14 (55.5% increase). These results demonstrate that functional 4-ASA-based co-SAMs critically enhance the UV resilience of Pb-Sn-based inorganic PSCs.

In addition, the long-term operational stability of unencapsulated NBG devices was evaluated following the ISOS-L-II protocol (continuous 1 sun illumination in  $\text{N}_2$  at room temperature) (Supplementary Fig. 42). The control device degraded markedly, retaining only 80% of its initial PCE after 600 hours, whereas the 4-ASA device retained over 90% after 1440 hours. Under thermal aging conditions (ISOS-L-2I; continuous one-sun illumination in  $\text{N}_2$  at 85 °C), the PCE decay profiles (Fig. 5g) reveal a clear enhancement in operational thermal stability for the 4-ASA-based device. While the control device retained only 73% of its initial PCE after 480 hours, the 4-ASA-based device preserved over 80% after 720 hours. To further substantiate the stability of the 4 T TSCs, the operational stabilities of the WBG Pb-Sn perovskite sub-cell and the integrated 4 T tandem device were evaluated under the same protocol (Supplementary Fig. 43). The unencapsulated WBG sub-cell retained 78.9% of its initial PCE, while the integrated 4 T tandem device maintained 80.1% of its initial efficiency, indicating favorable thermal stability. These results provide a strong foundation for the future development of stable Cs-based inorganic tandem systems, and further advances in encapsulation and interface engineering are expected to represent another significant step toward enhancing device stability. The stability improvement in NBG sub-cell is mainly attributed to two synergistic effects arising from the firmly bound 4-ASA-based co-SAMs (Supplementary Fig. 44). First, the  $-\text{NH}_2$  functional group in 4-ASA forms strong coordination bonds with undercoordinated  $\text{Pb}^{2+}/\text{Sn}^{2+}$  ions at the perovskite bottom surface, thereby passivating iodine vacancies (as evidenced by DFT results in Supplementary Fig. 25), and is expected to suppress the ionic diffusion pathway. Second, the co-SAMs monolayer serves as a physical barrier to reduce direct contact sites between  $\text{NiO}_x$  and the Pb-Sn perovskite. This effectively inhibits the notorious interfacial  $\text{Sn}^{2+}$  oxidation and suppresses the formation of interfacial defects that serve as nucleation sites for ionic redistribution. To elucidate suppressed ion accumulation, time-of-flight secondary ion mass spectrometry (TOF-SIMS) measurements were performed (Fig. 5h-k). The detection and spatial distribution of phosphorus (P) signals in the control device (Fig. 5h,i) indicate successful deposition of Me-4PACz at the  $\text{NiO}_x$ /perovskite interface. In comparison, co-localized P and sulfur (S) signals in the 4-ASA device (Fig. 5j,k) further suggest interfacial co-existence of Me-4PACz and tailored 4-ASA molecules. Notably, depth profiling reveals a substantially higher iodide ion (I<sup>-</sup>) concentration at the  $\text{NiO}_x$ /perovskite interface in the control device relative to the 4-ASA device, indicating ionic accumulation suppression by 4-ASA-based co-SAMs. Complementary three-dimensional (3D) I<sup>-</sup> mapping provides spatial resolution of this suppressed ionic diffusion throughout the modified device architecture. Furthermore, TOF-SIMS depth profiling was conducted on

devices after 240 h of operational stress under one-sun illumination. As presented in Supplementary Fig. 45, the control device shows pronounced iodide distribution at the  $\text{NiO}_x$ /perovskite interface after illumination, and the interfacial iodide enrichment indicates the occurrence of iodine accumulation. In contrast, the 4-ASA device shows substantially suppressed iodide enrichment at the  $\text{NiO}_x$ /perovskite interface while maintaining a uniform iodide distribution across the perovskite film following illumination. These spatially resolved measurements provide strong evidence that co-SAMs modification suppresses iodide redistribution under the tested conditions, thereby enhancing device operational stability.

In summary, we demonstrated a promising dipole engineering strategy via co-assembling azobenzene derivatives with Me-4PACz to modulate interfacial dipole orientation and strength, while concurrently mitigating SAMs aggregation, energy level mismatch, and  $\text{Sn}^{2+}$ -related defects at the  $\text{NiO}_x/\text{Pb-Sn}$  inorganic perovskite interface. Among the customized co-adsorbents, 4-ASA bearing a sulfonic acid anchoring group exhibited optimal performance. The combined effect of its intermolecular interaction with Me-4PACz and its strong dipole moment (8.8 D) resulted in uniform surface modification of  $\text{NiO}_x$ , minimizing the interface energy level offset to 0.02 eV. Additionally, via specific interactions with perovskite precursors, it improved perovskite film crystallinity, suppressed interfacial  $\text{Sn}^{2+}$  oxidation and ionic accumulation, while significantly reducing defect density. This synergistic effect enabled single-junction NBG Pb-Sn-based inorganic PSCs to achieve a record scanned PCE of 17.82%, with excellent stability under various conditions. Furthermore, by integrating with 1.74 eV WBG  $\text{CsPb}_{0.7}\text{Sn}_{0.3}\text{IBr}_2$  PSCs, we fabricated the first 4 T TSCs in which both the top and bottom subcells employ Cs-based inorganic Pb-Sn perovskite absorbers, yielding a total scanned PCE of 20.36%. Our study demonstrates a promising avenue for high-performance and stable Pb-Sn-based inorganic PSCs and their tandems. Future work could further optimize the assembly conditions and chemical structures of co-adsorbents to precisely control interfacial dipole orientation and strength, while extending their applications to flexible, large-area devices and other perovskite-based optoelectronic devices.

## Methods

### Materials

The ITO substrates, nickel oxide ( $\text{NiO}_x$ ), [6,6]-Phenyl C61 butyric acid methyl ester (PCBM), fullerene ( $\text{C}_{60}$ ), lead iodide ( $\text{PbI}_2$ ), and tin iodide ( $\text{SnI}_2$ ) were purchased from Advanced Electron Technology Co., Ltd. The 4-[(4-aminophenyl)diazanyl] benzoic acid (4-ACA), 4-[(4-aminophenyl)diazanyl] benzene sulfonic acid (4-ASA), manganese iodide ( $\text{MnI}_2$ ) and tin fluoride ( $\text{SnF}_2$ ) were purchased from Alfa Aesar. Lead bromide ( $\text{PbBr}_2$ ), cesium iodide (CsI), Methylammonium iodide (MAI), Methylammonium bromide (MABr), Phenethylammonium bromide (PEABr), and bathocuproine (BCP) were purchased from Xi'an Yuri Solar Co., Ltd. Tin bromide ( $\text{SnBr}_2$ , 99.9%), dimethyl sulfoxide (DMSO, 99.9%) and dimethylformamide (DMF, 99.8%) were purchased from Sigma-Aldrich. [4-(3,6-dimethyl-9H-carbazol-9-yl)butyl]phosphonic acid (Me-4PACz), 4-Aminophenylphosphonic acid, aniline, and other reagents were all purchased from the reagent platform Energy Chemical. Zinc oxide ( $\text{ZnO}$ ) nanoparticles were synthesized at room temperature by solution processing and dispersed in n-butanol and chloroform<sup>56</sup>.

### Synthesis of [4-[(4-aminophenyl)diazanyl]phenyl]phosphonic acid (4-APA)

To a 100 mL two-necked flask containing 4-aminophenylphosphonic acid (800 mg, 4.6 mmol) immersed in an ice-water bath, concentrated hydrochloric acid (1.2 mL) was added dropwise with stirring. Distilled water (15 mL) was then added to the mixture. In a separate beaker, sodium nitrite ( $\text{NaNO}_2$ , 700 mg, 10 mmol) was dissolved in distilled water (5 mL), and this solution was added dropwise to the reaction

flask maintained at 0 °C. The mixture was stirred for 1 hour. Subsequently, a solution of aniline (466 mg, 5 mmol) and sodium hydroxide (770 mg) in distilled water (12 mL) was prepared. After cooling to ambient temperature to complete dissolution, the solution was gradually introduced into the reaction flask. The reaction mixture pH was monitored using pH indicator paper and carefully adjusted to 5–7. Upon reaching this pH range, an immediate color change from colorless to orange was observed. The reaction was continued with stirring at 0 °C for an additional 2 hours. The resulting orange precipitate was collected and repeatedly washed with distilled water via centrifugation. Final purification was achieved by recrystallization from ethanol, affording the product 4-APA 650 mg, with a yield of 51%.

### Device fabrication

Narrow-bandgap perovskite solar cells (NBG PSCs): ITO glass substrates ( $7 \Omega \text{ sq}^{-1}$ ) were thoroughly cleaned and treated with plasma for 5 min before use. Then, a layer of  $\text{NiO}_x$  was spin-coated at 4000 rpm for 30 s and annealed at 120 °C for 10 min in ambient air. After cooling to room temperature, the substrates were transferred into a  $\text{N}_2$ -filled glovebox. For control devices, the Me-4PACz ( $0.5 \text{ mg mL}^{-1}$  in ethanol) was spin-coated on  $\text{NiO}_x$  layer at 4000 rpm for 30 s. After annealing at 100 °C for 10 min, the substrates were cooled down to room temperature and washed with IPA through spin-coating at 3000 rpm for 30 s, then annealed at 100 °C for 5 min. For 4-ACA, 4-APA, and 4-ASA devices, the composition of SAMs solution was changed to a mixture ( $0.5 \text{ mg Me-4PACz}$  and  $0.3 \text{ mg 4-ACA/4-APA/4-ASA}$  dissolved in 1 mL ethanol with a molar ratio of approximate 3:2). The  $\text{CsPb}_{0.6}\text{Sn}_{0.4}\text{I}_3$  inorganic perovskite films were spin-coated from a precursor solution containing CsI (0.9 M),  $\text{PbI}_2$  (0.54 M),  $\text{SnI}_2$  (0.36 M),  $\text{MnI}_2$  (0.036 M) and  $\text{SnF}_2$  (0.036 M) in DMF: DMSO = 3:1 in volume ratio, with a spin protocol of 4000 rpm for 70 s. To suppress the oxidation of  $\text{Sn}^{2+}$  induced by the DMSO,  $5 \text{ mg mL}^{-1}$  Dimethyl Ketoxime (DMKO) was adopted as reducing agent<sup>48</sup>. Besides, 200  $\mu\text{L}$  of MABr solution ( $5 \text{ mg mL}^{-1}$  in IPA) was quickly dripped onto the perovskite surface at 20 s before the end of protocol to regulate the crystallization of Pb-Sn perovskite. The as-prepared perovskite thin films were annealed at 70 °C for 2 min, and then at 120 °C for 10 min. After cooling to room temperature, 70  $\mu\text{L}$  of PEABr ( $2 \text{ mg mL}^{-1}$  in IPA:DMSO = 95:5 solution) was spin-coated onto the surface of perovskite films for surface defect passivation. Finally, C60 (25 nm), BCP (6 nm), and Ag (100 nm) were thermally evaporated.

Semi-transparent wide-bandgap perovskite solar cells (WBG PSCs):  $\text{NiO}_x$  layers were deposited the same as NBG PSCs. Then, a layer of 4-aminobenzenesulfonic acid (4-ABSA,  $2 \text{ mg mL}^{-1}$  in DI water) was spin-coated at 3000 rpm for 30 s and annealed at 100 °C for 10 min in ambient air. After cooling to room temperature, the substrates were transferred into a  $\text{N}_2$ -filled glovebox. The  $\text{CsPb}_{0.7}\text{Sn}_{0.3}\text{IBr}_2$  inorganic perovskite films were spin-coated from a precursor solution containing CsI (1 M),  $\text{PbBr}_2$  (0.7 M),  $\text{SnBr}_2$  (0.3 M), and  $\text{SnF}_2$  (0.03 M) in DMSO, with a spin protocol of 1000 rpm for 3 s and then at 4000 rpm for 30 s. Subsequently, the corresponding as-deposited perovskite films were treated by spin-coating 150  $\mu\text{L}$  of MAI solution ( $5 \text{ mg mL}^{-1}$  in IPA) at 4000 rpm for 15 s. After that, the films were annealed at 60 °C for 1 min and then at 160 °C for 10 min. After cooling to room temperature, 10  $\text{mg mL}^{-1}$  PCBM in chlorobenzene was spin-coated at 1000 rpm for 40 s, followed by annealing at 60 °C for 5 min. After that, a layer of ZnO was further deposited by spin-coating the prepared ZnO precursor at 3000 rpm for 30 s and then annealed at 150 °C for 10 min. Finally, 70 nm semitransparent IZO electrode was sputtered at a constant RF power of 150 W, Ar flow rate of 12 sccm, and the working pressure was 0.5 Pa.

### Characterization

**Film Properties.** CV measurements were conducted using a CHI 660E electrochemical workstation (CHI Instruments) with a custom-designed three-electrode glass cell. CV scans were recorded at a scan

rate of  $50 \text{ mV s}^{-1}$  over a potential window of  $-1 \text{ V}$  to  $+1 \text{ V}$ . Electrical resistivity measurements were performed on a Quantum Design PPMS-9 system using the Van der Pauw method at room temperature. The glass/ $\text{NiO}_x$ /co-SAMs films were cut into  $5 \times 5 \text{ mm}^2$  square pieces. Four ohmic contacts were applied at the four corners of each sample using a PPMS spring copper probe. The measurement was configured in in-plane geometry with current parallel to the film surface. The film thickness  $t$  (total thickness of  $\text{NiO}_x$ /co-SAM) was approximately 30 nm and was used to convert the sheet resistance  $R_s$  into volume resistivity according to  $\rho = R_s \cdot t$ . Twenty replicate measurements were performed for each sample, and the results are presented as mean  $\pm$  standard deviation. Ultraviolet photoelectron spectroscopy (UPS, ESCALAB 250Xi, Thermo Fisher) measurements were carried out using a He I discharge lamp (21.22 eV). The XRD patterns (2 $\theta$  scans) were obtained on Bruker Advanced D8 X-ray diffractometer using  $\text{Cu K}\alpha$  ( $\lambda = 0.154 \text{ nm}$ ) radiation. A UV-Vis spectrophotometer (Agilent Cary 5000) was used to collect the transmittance spectra of the substrates and absorbance spectra of the perovskite films. Steady-state photoluminescence (PL) spectra were recorded on a Shimadzu RF-5301pc. Time-resolved photoluminescence spectra were measured on a PL system (Fluo-Time 300) under excitation with a picosecond pulsed diode laser at 640 nm wavelength with a repetition frequency of 1 MHz. The morphology of the films was studied by field-emission scanning electron microscopy (FESEM, TESCAN, MIRA3) and atomic force microscopy (AFM, Bruker Dimension Icon). TOF-SIMS measurements were conducted using a ToF.SIMS 5-100 instrument (IONTOF GmbH), with the acquisition parameters set as: primary ion of  $\text{Bi}_3^+$ , acceleration energy of 30 keV, primary ion current of 0.75 pA, analysis raster size of  $50 \mu\text{m} \times 50 \mu\text{m}$ , and operated in high mass resolution mode; the sputtering parameters included a Cs ion gun as the sputter source, sputtering energy of 1 keV, sputter current of 60 nA, sputter raster area of  $200 \mu\text{m} \times 200 \mu\text{m}$ , and a sputtering rate of  $-0.84 \text{ nm s}^{-1}$  (calibrated on a GaN substrate).

**XPS characterization.** X-ray photoelectron spectroscopy (XPS) was conducted on a Thermo Scientific™ K-Alpha™ spectrometer equipped with a monochromatic Al  $\text{K}\alpha$  X-ray source (1486.6 eV) operating at 100 W. Samples were analyzed under high vacuum ( $P < 10^{-8} \text{ mbar}$ ) with a pass energy of 20 eV for high-resolution scans. 10 scans were collected for each spectrum to improve the signal-to-noise ratio and data reliability. To minimize surface contamination, all freshly cleaved samples were loaded into the XPS chamber within 2 hours of preparation. All binding energies were calibrated against the adventitious C 1s peak at 284.8 eV for charge correction. For spectral deconvolution, a Shirley background was adopted, and the full width at half maximum (FWHM) was kept consistent during peak fitting. Specifically, for Ni 2p<sub>3/2</sub> analysis, a Shirley background with 9 iterations was employed, and the fitting range was strictly constrained to 845–868.5 eV. All main peaks were modeled using GL(35) mixed line shapes (35% Lorentzian), with fixed binding energies of  $\text{Ni}^{2+} = 853.2 \text{ eV}$ ,  $\text{Ni}^{3+} = 855.0 \text{ eV}$ , and  $\text{Ni}^{4+} = 856.5 \text{ eV}$ , allowing only a fine adjustment within  $\pm 0.2 \text{ eV}$ . The FWHM of the three valence states was uniformly fixed at 1.8 eV. During fitting, only the peak areas were optimized, while the peak-shape parameters were kept constant. For uncertainty analysis, the same Ni 2p<sub>3/2</sub> spectrum was independently fitted three times by varying the initial fitting parameters within reasonable ranges (binding energy:  $\pm 0.1 \text{ eV}$ ; FWHM:  $\pm 0.1 \text{ eV}$ ) to evaluate the robustness of the valence-state quantification.

**Device Performance.** The current density-voltage (J-V) characteristics of the devices were collected using a source meter (Keysight B2901A) coupled with a solar simulator (Enlitech SS-F5-3A), following a measurement protocol consisting of forward scans ( $-0.1 \text{ V}$  to  $1.2 \text{ V}$ ) and reverse scans ( $1.2 \text{ V}$  to  $-0.1 \text{ V}$ ), using 20 mV voltage steps and 10 ms delay intervals. The light intensity was calibrated to AM 1.5 G

(100 mW cm<sup>-2</sup>) by using a reference Si solar cell. All measurements were performed in a nitrogen-filled glovebox at room temperature using a laser-cut metal mask, which defined active areas of 0.04 cm<sup>2</sup> for small-area champion devices and 1 cm<sup>2</sup> for area-scaled devices. The external quantum efficiency (EQE) spectra were recorded with a quantum efficiency measurement system (Enlitech QER-3011) in which the light intensity at every wavelength was calibrated with a Si detector before measurement. The maximum-power point (MPP) output characteristics were evaluated through tracking the steady-state current density at the MPP voltage over a 10-minute duration, with data recorded at 2 s intervals under a controlled temperature of 25 °C in a nitrogen-filled glovebox to eliminate ambient degradation. The 4T tandem efficiency and EQE were obtained by summing the measured efficiency and EQE of the standalone WBG top cell and that of the filtered NBG bottom cell. The standalone WBG top cell was measured directly under full-spectrum AM 1.5 G light without any optical filter. For filtered NBG bottom cell testing, the fully fabricated semi-transparent WBG top cell was physically placed atop the NBG device to filter incident sunlight, thereby reproducing the optical environment experienced by the bottom cell in the assembled 4T tandem configuration. No spectral mismatch correction was applied during the measurements.

**Charge dynamics.** Hole-only devices with a configuration of ITO/NiO<sub>x</sub>/Co-SAMs/Perovskite/PTAA/Ag were fabricated for SCLC measurement. The J-V responses of the hole-only devices were measured at room temperature in the dark. Transient photovoltage photocurrent decay measurements were conducted on a home-made system. A white light bias on the sample was generated from an array of diodes. Red light pulse diodes (0.05 s square pulse width, 100 ns rise and fall time) controlled by a fast solid-state switch were used as the perturbation source. The transient photocurrent was measured using 20 Ω external series resistance to operate the device in short circuit. Similarly, transient photovoltage was measured using 1 MΩ external series resistance to operate the device in open circuit. The voltage dynamics on the resistors were recorded on a digital oscilloscope (Tektronix MD03032). The perturbation red light source was set to a suitably low level to the white diode array with light intensity equivalent to 100 mW cm<sup>-2</sup> of a standard solar simulator. Mott-Schottky analyses were carried out on an electrochemical workstation (IM6eX, Zahner, Germany) in a voltage range of -0.1 to 1.1 V under dark conditions.

### Computational details

**DFT calculations:** All first-principles calculations based on density functional theory (DFT) were performed using the Vienna Ab initio Simulation Package (VASP)<sup>57</sup>, with the projector augmented wave (PAW) method employed<sup>58–60</sup>. The Perdew–Burke–Ernzerhof (PBE) functional within the generalized gradient approximation (GGA) was adopted to describe the electron exchange and correlation effects<sup>61</sup>. The van der Waals interactions were accounted for using the Grimme DFT-D3 method with the zero-damping function<sup>62</sup>. For adsorption energy, binding energy, and defect passivation calculations, a plane-wave cutoff energy of 400 eV was employed, with energy and force convergence criteria set to 1×10<sup>-4</sup> eV and 0.05 eV Å<sup>-1</sup>, respectively. A 4 × 3 × 1 supercell of NiO (100) was used to model the NiO<sub>x</sub> layer, a 3 × 3 × 2 supercell of CsPb<sub>0.6</sub>Sn<sub>0.4</sub>I<sub>3</sub> (100) was used for the perovskite layer, and a vacuum spacing greater than 20 Å was included in all slab models. Additionally, the electrostatic potential (ESP) of Me-4PACz, 4-ACA, 4-APA, and 4-ASA was computed using all-electron double-ζ valence basis sets. All crystal structures were constructed using Materials Studio and visualized with VESTA<sup>63</sup>.

**MD simulation:** Classical molecular dynamics (MD) simulations were performed using the LAMMPS package to investigate the adsorption behavior of co-SAMs on the NiO<sub>x</sub> surface<sup>64</sup>. A NiO (100) slab was constructed based on a 98.4 Å × 98.4 Å × 6 Å supercell. On top of the

surface, the Packmol program was employed to generate co-SAM layers<sup>65</sup>. Four adsorption models were considered: (i) 150 Me-4PACz molecules, (ii) 150 Me-4PACz and 100 4-ACA molecules, (iii) 150 Me-4PACz and 100 4-APA molecules, and (iv) 150 Me-4PACz and 100 4-ASA molecules. The ratios between Me-4PACz and the tailored molecules were set to match the experimental molar ratios (3:2). Intermolecular interactions of Me-4PACz, 4-ACA, 4-APA, and 4-ASA molecules were described using the OPLS-AA force field<sup>66</sup>, while the atomic interactions within the NiO<sub>x</sub> substrate were modeled using a Buckingham potential<sup>67</sup>. The interactions between the molecules and surface were described by combining the force field parameters using geometric mixing rules (Lorentz–Berthelot)<sup>68</sup>. The interfacial systems were equilibrated in the canonical NVT ensemble at 300 K using the Nosé–Hoover thermostat for 1 ns<sup>69,70</sup>. Atomic motions were integrated using the velocity-Verlet algorithm with a 1 fs time step. Periodic boundary conditions were applied in the x and y directions, while the NiO<sub>x</sub> substrate was fixed. Surface adsorption analysis was conducted using an in-house analysis code. The surface coverage was evaluated by identifying atoms of molecules located above the NiO<sub>x</sub> surface within a distance threshold of 3 Å and a radial threshold of 2 Å.

### Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

### Data availability

The main data supporting the findings of this study are available within the published article and its Supplementary Information and source data files. Additional data are available from the corresponding author on request. Source data are provided with this paper.

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## Author contributions

Weihai Zhang, H.W. and A.A. conceived the project. Weihai Zhang and Heng Liu were mainly responsible for fabrication of high-performance Pb-Sn inorganic perovskite solar cells and tandem devices. Haiyan Li and J.M. were mainly responsible for DFT calculations. X.Z. was mainly responsible for materials synthesis and characterization. F.Y. and J.D. carried out MD simulations. T.S. revised the DFT calculations and the MD simulations. Weihai Zhang, X.Z., Heng Liu, Hui Li, Wenhao Zhou, and Wenjun Zhang conducted the material and device characterizations. T.S., H.W. and A.A. provided valuable suggestions for the manuscript. Weihai Zhang wrote and revised the manuscript. Weihai Zhang, Haiyan Li, X.Z. and Heng Liu contributed equally to this work. All authors discussed the results and commented on the manuscript.

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## Competing interests

The authors declare no competing interests.

## Additional information

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